

CSE211 – Computer Organization & Design

WORKSHEET – 2

Subjective Questions (3)

1. Describe stack organization and explain how PUSH and POP operations work.
2. Explain different types of addressing modes.
3. Explain the concept of general register organization.

MCQ based questions (15)

1. A function call stores return addresses in a LIFO structure.
Which organization is being used?
 - A. Queue
 - B. Stack
 - C. Register file
 - D. Cache
2. A control unit selects specific registers for ALU operations using control signals.
What does this selection enable?
 - A. Execution of micro-operations
 - B. Operations in memory expansion
 - C. Permanent data storage operations
 - D. Input-output operations
3. An instruction accesses data directly from a specified memory address.
Which addressing mode is used?
 - A. Immediate
 - B. Indirect
 - C. Direct
 - D. Indexed
4. An instruction contains the actual operand value within itself.
How is this mode interpreted?
 - A. Operand is in memory
 - B. Operand is in register
 - C. Operand is in cache
 - D. Operand is part of instruction
5. An array element is accessed using base address plus index register.
What is the role of the index register?
 - A. Stores opcode
 - B. Provides offset value
 - C. Controls interrupts

D. Stores I/O data

6. A processor uses simple instructions that execute in one clock cycle.

Which architecture does this describe?

- A. CISC
- B. Microprogrammed
- C. Stack-based
- D. RISC

7. A processor executes complex instructions like string manipulation in a single instruction.

What does this indicate?

- A. CISC design
- B. RISC design
- C. Stack organization
- D. Pipeline hazard

8. An instruction contains fields for opcode, register, and address.

What is this structure called?

- A. Addressing mode
- B. Micro-operation
- C. Control word
- D. Instruction format

9. Data is transferred asynchronously between two devices without a shared clock.

Which scheme is used?

- A. Synchronous transfer
- B. Asynchronous transfer
- C. Interrupt-driven
- D. DMA only

10. A device transfers data directly to memory without CPU involvement.

What is this technique?

- A. Interrupt handling
- B. Programmed I/O
- C. DMA
- D. Polling

11. Multiple devices request service simultaneously, and CPU selects one based on priority.

What concept is applied?

- A. Stack organization
- B. Instruction format execution
- C. Addressing mode
- D. Interrupt priority system

12. After an ALU operation, flags like zero and carry are updated.

Where are these flags stored?

- A. PSW
- B. PC
- C. MAR
- D. IR

13. A zero flag is set after an operation.

What does this indicate?

- A. Result is negative
- B. Result is zero
- C. Overflow occurred
- D. Interrupt triggered

14. Operands are pushed into a stack and operation uses top elements.

How are operands accessed?

- A. By address field
- B. By register selection
- C. By top of stack
- D. By memory indexing

15. A processor uses registers for operations, supports conditional branching, handles interrupts, and updates flags after execution.

Which combination of concepts is involved?

- A. Only data transfer, address, data, control
- B. Only stack operations, top of stack
- C. Addressing modes, immediate, indexed and relative addressing
- D. Register organization, program control, interrupts, and PSW